BENEDICT ARMSTRONG

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Education

- **ETH Zürich**
  M.Sc. Computational Science and Engineering, Robotics track
  2026 — present
- **ETH Zürich**
  B.Sc. Computational Science and Engineering, Robotics track
  2021 — 2025
- **Vienna University of Technology**
  B.Sc. Computer Science, Software & Information Engineering track
  2019 — 2021 (switch to ETH)

Research

- **AI for Gravitational-Wave Physics**, ML Research Intern
  Computational Applied Mathematics & AI Lab (CAMAIL), ELLIS Institute Tübingen and Max Planck Institute for Intelligent Systems
  In collaboration with the LIGO Lab at MIT · supervised by T. Konstantin Rusch
  June 2025 — Summer 2026

  When two neutron stars spiral into each other, they send ripples through spacetime that LIGO measures as length changes far smaller than a proton. These signals can last minutes and are buried in non-stationary, glitchy detector noise, while astronomers need an alert within seconds to point telescopes at the merger. I work on Linear Oscillatory State-Space Models (LinOSS): sequence models that handle very long inputs efficiently and make it possible to detect binary neutron star mergers in real time.
  - Designed and trained state-space models for low-latency binary neutron star detection on LIGO strain data.
  - Built the training and evaluation pipeline in PyTorch (Lightning) and JAX/Equinox.
  - Co-developed PhyTS, a benchmark of physics time-series tasks (NeurIPS 2026).

Talks

- *Linear Oscillatory State-Space Models for Binary Neutron Star Detection*
  Benedict Armstrong · AI for Gravitational Waves Workshop, CERN, Geneva, May 2026
- *Direct Parameter Estimation of Binary Neutron Star Mergers with State Space Models*
  Kevin (Kyungseop) Yoon · Fast Machine Learning for Science Conference 2026, UC San Diego, September 2026



Born and raised in Vienna, currently living in Zürich. In my free time, I like to code, tinker with electronics, design and make furniture, and generally enjoy all types of DIY/maker projects. I am also interested in FPV drones, IoT prototyping, 3D printing, tennis, cycling, triathlons, and of course, all things computer science.

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Projects

Presio · presio.ch

Web app for presenting PDF slides with a presenter controller and a mirrored audience window. PDFs stay in the browser by default; signed-in users can share a session code so others can follow live on their own devices.

AcceptOdds · acceptodds.com

Short description of what AcceptOdds does, who it is for and what is technically interesting about it.

Skills

- English, German
- Python, PyTorch, JAX, C++, C, Java, JavaScript, PHP, SQL, Angular, React, NextJS, Svelte(Kit), FastAPI, REST, HTML, CSS, Mongo, Postgres, Docker, Git, Arduino/ESP8266/32 based hardware development, Scrum

Publications

- **PhyTS: A Benchmark for Scientific Time Series**
  Benedict Armstrong, Jeroen Audenaert, Hannah P. Binney,
  … , Kyungseop Yoon (26 authors)
  NeurIPS 2026 , Sydney

Experience

- **Freelance Software Engineer**, Armstrong Systems 2022 — 2024

  Founded Armstrong Systems, software consulting for clients in Zürich. Full-stack development of various projects including a web-based data management tool, visualization dashboards and large-scale data ingestion pipelines.
  - Scoped, built and maintained a web-based data management tool used day to day by the client's team.
  - Built interactive visualization dashboards on top of the clients' operational data.
  - Designed large-scale data ingestion pipelines, from source integration to deployment and monitoring.

- **Coding Assistant for LehrFEM++ Finite Element Library**, Seminar for Applied Mathematics, ETH Zürich Fall 2024

  Coding assistant for the C++ finite element library LehrFEM++, used in the course *Numerical Methods for Partial Differential Equations* taught by Prof. Dr. Ralf Hiptmair.
  - Extended and maintained the LehrFEM++ codebase used by students in the course's programming exercises.
  - Wrote documentation and examples to make the library easier for students to pick up.

- **Paramedic (civil service)**, Samariterbund 2018 — 2019

  Worked as a paramedic in Vienna as part of my civil service.

- **Various summer jobs**, Various Summers 2017–2021

  Various summer jobs in Vienna, including working as a postman, food delivery driver and math/CS tutor for high school students.

Teaching

Department of Computer Science, ETH Zürich

- **Teaching Assistant**, *Software Engineering Fundamentals* Spring Semester 2024
  Teaching Assistant for the course Software Engineering Fundamentals.
- **Head Teaching Assistant**, *Computer Science II* August 2023 — February 2024
  I was responsible for the organization of the teaching assistants and the exercise classes. In addition I worked closely with the lecturers preparing exercise material and made sure the course ran smoothly.
- **Teaching Assistant**, *Computer Science II* August 2022 — February 2023
  I taught one of the exercise classes and reviewed student submissions. The course provides the foundations for the design and analysis of algorithms, from sorting to graph problems.

Research Group Industrial Software (INSO), TU Wien

- **Teaching Assistant and Full Stack Developer**, *Security for Systems Engineering* Spring 2021
  Together with a small team we developed security challenges for teams of students to compete in a Capture the Flag (CTF) competition. In addition I supervised students during the competition and helped administer exams later in the semester.